

Ligand Control in Enantioselective Desymmetrization of Bicyclic Hydrazines: Rhodium(I)-Catalyzed Ring-Opening *versus* Hydroarylation

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Abstract: The efficient desymmetrization of 2,3-bicyclic hydrazines with boronic acids through rhodium-catalyzed ring-opening or reductive arylation is described. Excellent levels of enantioselectivity are achieved in ring-opening with *ortho*-substituted boronic acids, using Josiphos-type ligands. Alternatively, reductive arylation occurs selectively with electron-

poor Josiphos and Walphos ligands. A C–H activation/1,4-migration mechanism was established through deuterium transfer experiments.

Keywords: asymmetric catalysis; C–H activation; desymmetrization; ferrocene ligands; rhodium

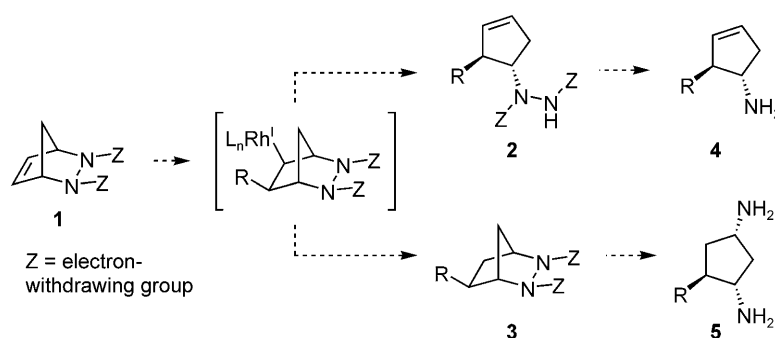
Introduction

The development of new enantioselective reactions is a formidable challenge in synthetic chemistry. Desymmetrization of simple *meso* compounds is an efficient way of generating chiral building blocks with one or more stereocenters.^[1] For instance, bicyclic compounds including oxa- and azabicycles can be desymmetrized under transition metal catalysis to yield stereochemically rich scaffolds and synthons.^[2] Related systems such as 1,2-diazabicycles **1**, have received considerably less attention. Our previous success in ring-opening reactions prompted us to investigate de-

symmetrization of **1** ultimately leading to two enantioselective transformations as described below.^[3,4]

Compound **1** can undergo ring-opening to yield cyclopentene **2** (Scheme 1). Alternatively, nucleophiles can be added reductively to **1** to give bicycle **3**. Reduction and deprotection of compounds **2** and **3** can yield **4** and **5**, respectively, with retention of the relative stereochemistry. Such substituted chiral scaffolds are common in biologically significant molecules, and their stereoselective synthesis is a worthwhile goal (Figure 1).^[5–7]

Our interest in the selective desymmetrization of **1** is preceded by the work of a number of groups. Most



Scheme 1. Divergent pathways in the desymmetrization of diazabicycles, leading to chiral substituted cyclopentenones and cyclopentanes.

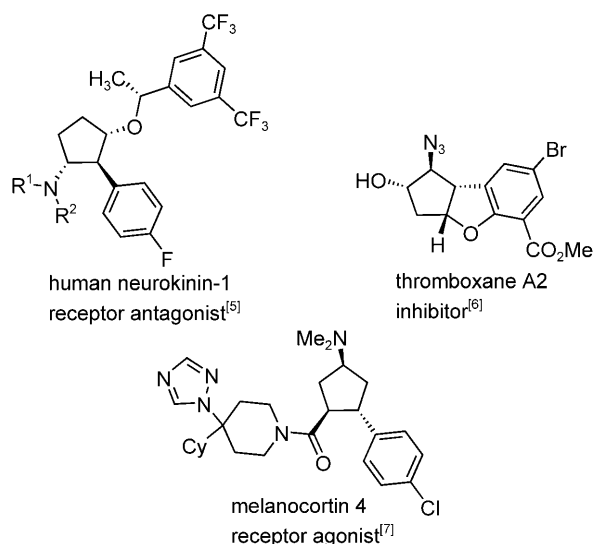


Figure 1. Biologically active compounds containing a substituted chiral cyclopentanamine core.

notably, Pineschi and co-workers reported the first asymmetric ring-opening of **1** with carbon nucleophiles under copper and rhodium catalysis.^[8] They established that Rh(I) catalysts promote the addition of a range of organoboron nucleophiles, although stereoinduction remained an area in need of further improvement.

A palladium-catalyzed ring-opening of **1** was reported by Radhakrishnan and co-workers, where the nucleophiles used included organostannanes and organoindiums.^[9] Similarly, Kauffman showed that a palladium-catalyzed coupling of aryl iodides and **1** was possible, yielding mixtures of **2** and **3**.^[10] To the best of our knowledge, an enantioselective route to **3** has not been reported.

Herein, we report our progress on the development of ligand controlled reactions where either **2** or **3** can be obtained. We describe the reaction parameters and requirements for a highly selective ring-opening of **1**, particularly with *ortho*-substituted boronic acids. In addition we have identified ligands and nucleophiles favouring what is formally a hydroarylation to **3**.^[3,11] We propose a 1,4-migration/C–H insertion mechanism based on deuterium quench experiments. Such a process differs from the proposed mechanisms for reductive arylation under palladium catalysis and thus opens up exciting synthetic opportunities.^[12]

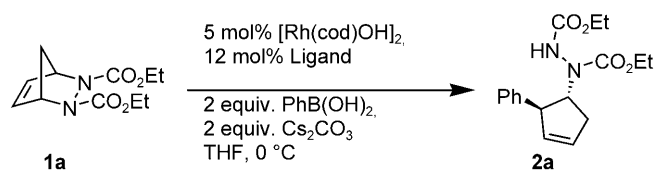
Results and Discussion

We elected to examine the desymmetrization of **1** using rhodium catalysts and boronic acids as nucleophiles. Rhodium complexes (i.e., [Rh(cod)OH]₂) were appealing because of their predictable reactivity in

asymmetric additions to bicyclic alkenes. Boronic acids are ideal carbon nucleophiles resulting from their high tolerance to air and moisture, and their easy accessibility. The [2,2,1]-diazabicycles **1** were conveniently prepared using Diels–Alder chemistry.^[13]

Enantioselective Ring-Opening

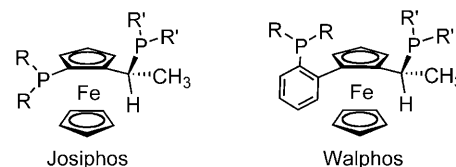
Our previous conditions for desymmetrizing *meso* allylic biscarbonates provided a starting point for enantioselective ring-opening of **1**.^[14] Preliminary results were promising (Scheme 2). Xylyl-P-phos^[15] generat-



Xylyl-P-phos	18% (87% ee)
<i>t</i> -Bu-Josiphos (J1)	72% (75% ee)

Scheme 2. Preliminary findings in the enantioselective ring-opening reaction.

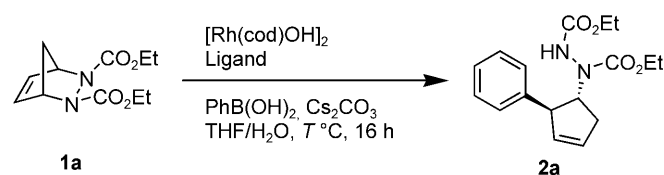
ed encouraging enantioselectivity (87% ee) and *t*-Bu-Josiphos (**J1**, Figure 2)^[16] showed high conversion (72% yield). A strong ligand-dependence on reactivity was observed (Table 1).



Ligand	R	R'
J1	phenyl	<i>t</i> -butyl
J2	cyclohexyl	<i>t</i> -butyl
J3	α -furyl	<i>t</i> -butyl
J4	<i>m,m'</i> -bis(trifluoromethyl)phenyl	<i>m,m'</i> -xylyl
W1	phenyl	<i>m,m'</i> -bis(trifluoromethyl)phenyl
W2	<i>m,m'</i> -dimethyl- <i>p</i> -methoxyphenyl	<i>m,m'</i> -bis(trifluoromethyl)phenyl
W3	phenyl	cyclohexyl

Figure 2. Structures of the Josiphos and Walphos ligands used in this study

A range of ligands was examined using phenylboronic acid. Good levels of enantioselectivity could be obtained with a subset of ligands, such as P-phos^[15] and Difluorphos^[17] (entries 1–4, Table 1). BINAP and its derivatives showed slightly lower selectivity, but marginally superior conversion (entries 5–8). Additionally, ferrocenyl phosphine ligand **J1** (*t*-Bu-Josiphos) showed high conversion and modest enantiose-

Table 1. Ligand screening for rhodium-catalyzed ring-opening of diazabicycles.^[a]

Entry	Ligand	Temp. [°C]	Yield [%] ^[b]	ee [%] ^[c]
1	Xylyl-P-phos	60	18	87
2	Xylyl-P-phos	25	–	n.d.
3	P-phos	25	–	79
4	Difluorophos	25	–	93
5	BINAP	60	15	n.d.
6	BINAP	25	6	70
7	Tolyl-BINAP	25	–	80
8	Xylyl-BINAP	0	–	84
9	SegPhos	25	–	86
10	<i>t</i> -Bu-Josiphos (J1)	25	78	69
11	<i>t</i> -Bu-Josiphos (J1)	0	80	77
12 ^[d,e]	J2	25	84	77
13 ^[e,f]	J4	25	10	50

^[a] Conditions: $[\text{Rh}]$ (5 mol%), ligand (12 mol%) and base (2 equiv.) were premixed in THF/water (50:1). Solution of **1** and $\text{PhB}(\text{OH})_2$ (2 equiv.) was added.

^[b] Yields of isolated products. “–” indicates that low conversion (< 10%) was observed by TLC.

^[c] Enantiomeric excess determined by chiral HPLC.

^[d] Reaction performed using **1d**.

^[e] See Figure 2 for ligand structure.

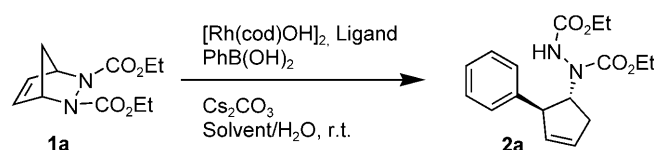
^[f] Compound **3** was a major side product.

lectivity (entries 10–13). Additional reaction parameters needed to be optimized since we were unable to attain both high conversion and enantioselectivity with any of the ligands. Importantly, a by-product was observed during ligand screening and was later identified as the reductive arylation product **3** (*vide infra*).

Solvent effects were examined for the best ligands, where THF and toluene proved to be the most adequate solvents (entries 8–10, Table 2). A preliminary screen of boronic acids showed that *t*-Bu-josiphos (**J1**) afforded excellent stereoselection with *ortho*-substituted boronic acids.^[3] Unfortunately, although Difluorophos afforded the highest *ee*, the reaction proceeded with very low conversion (entries 16–18).

Neither changes in concentration nor temperature influenced the product distribution. Below 0 °C the reaction was sluggish giving partial conversion when diazabicyclic **1d** was used (entry 12, Table 2).

Water was required to obtain a good yield. Indeed the reaction gave low conversion under anhydrous conditions (entry 11, Table 2). It is presumed that water is essential for regeneration of the catalyst. The chloride rhodium complex, for instance, did not give product without base and water, suggesting *in situ* for-

Table 2. Influence of solvent on the ring-opening reaction with selected ligands.^[a]

Entry	Ligand	Solvent	Yield [%] ^[b]	ee [%] ^[c]
1 ^[d]	BINAP	THF	6	70
2	BINAP	THF	16	71
3	BINAP	Dioxane	35	70
4	BINAP	Chloroform	38	60
5	BINAP	DCE	6	39
6	BINAP	Benzene	70	73
7	BINAP	Toluene	57	70
8 ^[e]	BINAP	Toluene	91	78
9 ^[e]	<i>t</i> -Bu-Josiphos (J1)	Toluene	93	76
10 ^[e]	<i>t</i> -Bu-Josiphos (J1)	THF	80	77
11 ^[d,f]	<i>t</i> -Bu-Josiphos (J1)	THF	29	67
12 ^[e,f]	<i>t</i> -Bu-Josiphos (J1)	THF	55	67
13 ^[f,g]	<i>t</i> -Bu-Josiphos (J1)	THF	79	51
14	<i>t</i> -Bu-Josiphos (J1)	dioxane	69	70
15	<i>t</i> -Bu-Josiphos (J1)	DME	Trace	66
16 ^[d]	Difluorophos	THF	Trace	93
17	Difluorophos	THF	Trace	80
18	Difluorophos	Toluene	Trace	78

^[a] Conditions as in Table 1, for more screening see Supplementary Information.

^[b] Yield of isolated products.

^[c] Enantiomeric excess determined by chiral HPLC.

^[d] Reaction performed without addition of water.

^[e] Reaction performed at 0 °C.

^[f] Starting material **1d** used (Boc protecting group).

^[g] Reaction performed at 50 °C.

mation of $[\text{Rh}(\text{cod})\text{OH}]_2$ (entries 7 and 8, Table 3). Although the reaction proceeded with excellent conversion in ethereal solvents, higher yield of **2a** could be obtained in toluene and water (entries 9 and 10, Table 2).

The nature of the protecting group on the diazabicyclic influenced the yield and selectivity (Table 3). The enantioselectivity was highest with Cbz-substituted derivative **1b** (entry 4). Phthalyl-substituted derivative **1c** and ethyl carboxylate-protected **1a** afforded **2** in high yield and moderate selectivity (entries 1–3 and 5). The increase in yield suggests that electron-poor protecting groups may improve the leaving group ability of the hydrazide, thereby increasing the yield and preventing a competing 1,4-migration from taking place (*vide infra*). With a Boc-protected derivative **1d**, the enantioselectivity and yield were slightly lower (entry 6).

In an attempt to extend the reaction to alternative organometallic nucleophiles, phenylmagnesium bromide and phenylzinc bromide were tested (entries 9

Table 3. Influence of protecting group.^[a]

Entry	1	Protecting group (Z)	Ph-B(OH) ₂	Yield [%] ^[b]	ee [%] ^[c]
1	1a	CO ₂ Et	Ph-	80	77
2 ^[d]	1a	CO ₂ Et	Ph-	96	71
3 ^[e]	1a	CO ₂ Et	Ph-	72	75
4	1b	CO ₂ Bn	Ph-	72	80
5	1c	Phthalyl	Ph-	99	70
6	1d	CO ₂ - <i>t</i> -Bu	Ph-	85	66
7 ^[e,f]	1d	CO ₂ - <i>t</i> -Bu	Ph-	0	—
8 ^[f]	1d	CO ₂ - <i>t</i> -Bu	Ph-	65	69
9 ^[e,g]	1d	CO ₂ - <i>t</i> -Bu	Ph-MgBr	65	12
10 ^[e,g]	1d	CO ₂ - <i>t</i> -Bu	Ph-ZnBr	63	22
11 ^[h]	1d	CO ₂ - <i>t</i> -Bu	Ph-	10	42
12 ^[i]	1d	CO ₂ - <i>t</i> -Bu	Ph-	84	1

^[a] Reaction conditions as described in Table 1.^[b] Yield of isolated products.^[c] Enantiomeric excess determined by chiral HPLC.^[d] Degassed solvents used.^[e] No base used.^[f] Reaction performed using [Rh(cod)Cl]₂ as rhodium source.^[g] Reaction performed under anhydrous conditions.^[h] Reaction conditions: 10 mol% Pd(CH₃CN)₂Cl₂ and 11 mol% Tol-BINAP in MeOH.^[18]^[i] Reaction conditions: 10 mol% Pd(CH₃CN)Cl₂ and 11 mol% **J1** in MeOH.^[18]

and 10, Table 3). Intriguingly, in spite of obtaining comparable yields, nearly racemic product was obtained. Clearly, the increased reactivity of the nucleophile or the presence of bromide salts was responsible for the dramatic difference.

In addition to rhodium catalysts, we tested palladium under conditions previously developed in ring-opening of bicyclic compounds (entries 11 and 12, Table 3).^[18] The reaction proceeded to give yields similar to rhodium, in analogy with Rhadakrishnan's results (entry 12).^[9] However, the enantiomeric excess of the products was low.

Scope of Boronic Acids in Ring-Opening of **1d**

The ring-opening reaction was shown to proceed with high conversion with a variety of arylboronic acids (Table 4). The yield depended largely on electronic changes in the nucleophile, where electron-rich boronic acids gave the highest yields (entries 5–7, 9, 10, 13 and 14). Decreased yield of ring-opening product may be caused by competing C–H insertion leading

Table 4. Ring-opening of diazabicycles with substituted boronic acids.^[a]

1d			6 – 18	
Entry	Product	Ar	Yield [%] ^[b]	ee [%] ^[c]
1 ^[d]	6		53	99
2	7		66	51
3 ^[e]	7		86	82
4 ^[d]	8		40	99
5	9		52	97
6 ^[f]	9		99	97
7	10		75	99
8 ^[d]	11		54	> 99
9	12		96	99
10	13		91	96
11 ^[d]	14		49	84
12	15		58	86
13	16		80	50
14	17		83	57
15	18		99	44

^[a] Reaction conditions as described in Table 1.^[b] Yield of isolated products.^[c] Enantiomeric excess determined by chiral HPLC.^[d] Reductive arylation product observed.^[e] Reaction performed using **W1** as ligand.^[f] Reaction performed in toluene/THF/H₂O mixture (1:1:0.1).

to product **3**, a mechanism that appears to be more facile with electron-deficient nucleophiles (entries 1–4, 8, 11, 12, *vide infra*). The product ratio could be influenced to favour ring-opening by performing the reaction in a toluene/THF solvent system (entry 6).^[19]

ortho-Substituted boronic acids were ideal in this transformation, giving products with excellent enantioselectivity (entries 1, 4–10). The nature of the *ortho* substituent could be varied extensively, giving good yields with electron-rich or neutral groups. It is apparent that steric bulk is important for generating good *ee*. Reaction conditions were mild and compatible with various functional groups (entries 10, 12 and 14). Styrylboronic acid reacted quantitatively, although selectivity was moderate (entry 15).

The stereochemical assignment of the substituted cyclopentenes **2** was difficult because of the presence of rotamers by NMR analysis. Fortunately, the relative stereochemistry could be determined for phthalyl-substituted **2c**, through X-ray crystallography.^[20] The absolute and relative stereochemistry of Boc-protected **2d** was determined by reduction of the N–N bond, deprotection to the free amine and comparison of the optical rotation to literature values.^[3]

Optimization of Reductive Arylation Reaction

During the course of the study we observed that the ratio of **2** and **3** varied as a function of the nucleophile and solvent. While **3** was initially a minor by-product, the discovery of sequential addition/C–H activation was intriguing and potentially applicable to more sophisticated reactions and therefore warranted a detailed study.

We observed that a substantial amount of **3** formed with heteroaromatic and electron-deficient boronic acids (entries 1, 4, 11, Table 4).^[21] Upon further investigation of the reaction with pyrimidine derivative **19** in the presence of D₂O, we observed deuterium incorporation at the *ortho* aromatic position (Scheme 3).

Formation of this product was thus hypothesized to occur *via* a 1,4-metal migration, which is a known process in rhodium catalysis.^[22] We speculate that electron-deficient boronic acids produce more reductive arylation because of increased acidity of the *ortho* C–H bond, which may facilitate oxidative insertion of rhodium(I). In accord with this reasoning, nearly ex-

clusive 1,4-migration was observed with thienylboronic acid to give **22** in 87% yield.

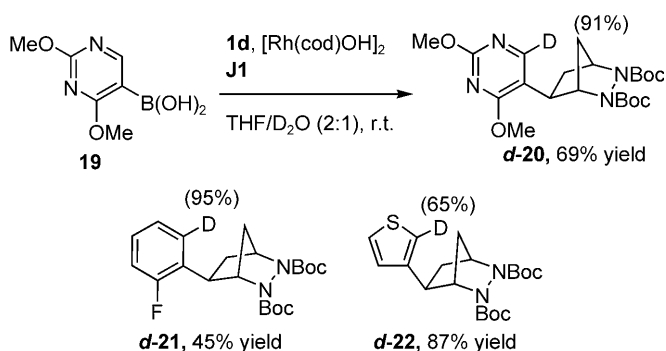
Since product **3** was initially observed during ligand screening we examined the influence of ligand on the **2**:**3** product ratio. A trend was noted between this ratio and the electronic density of the phosphorus atoms in the ligand (Table 5). Josiphos ligands bearing electron-poor aromatic phosphines produced more reductive arylation product **3d** (**J3** and **J4**, entries 3 and 5). Conversely, electron-rich ligands gave mainly ring-opening product **2d** (**J1** and **J2**, entries 1 and 2). Upon examination of a related Walphos family of ligands, multiple 1,4-migration was observed with **W1** to give the by-product **23** (entry 6, Scheme 4). Formation of **23** can be explained by carborhodation/1,4-migration followed by a reaction with a second equivalent of diazabicyclo, akin to the *merry-go-round* seen with norbornene under similar conditions.^[23]

Formation of other disubstituted compounds, **24** and **25**, occurred with 3-furylboronic acid and with Boc-protected 3-pyrrolylboronic acid. Compound **24** formed through a C–H insertion followed by ring-opening of a second equivalent of bicycle **1d**. Formation of **23** could be decreased by slow addition of **1d** (entry 7, Table 5). Overall, Walphos ligands were better than Josiphos ligands at inducing 1,4-migration (entries 2 and 4). This observation suggests that the bite angle, along with electronics, is an important factor in the observed chemoselectivity.

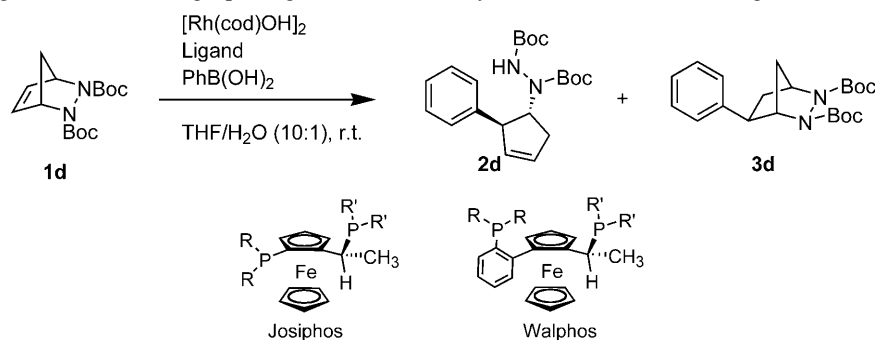
The chemoselectivity of the reaction was somewhat influenced by the solvent. Whereas the reductive arylation product was readily observed in THF (~20% yield), it either disappeared or diminished significantly in reactions run in a THF/toluene mixture (Table 4).^[19] Such an outcome suggested that coordinating solvents may favour rhodium C–H insertion by stabilizing a Rh(III) complex, while non-coordinating solvents favour ring-opening.

A number of reductive arylation products were synthesized to demonstrate the reaction's generality (Table 6). Under the current conditions, the yields and enantioselectivity are still variable. It appears that enantioselectivity with Walphos ligands is governed by factors different than those for Josiphos. Although not ideal, the moderate selectivity is encouraging.

As extensive C–H insertion was observed with both 3-thienylboronic acid and 3-furylboronic acid, we anticipated that 3-pyrrolylboronic acid would be an appropriate candidate to study the electronic influence of the aryl moiety by varying the nitrogen protecting groups. In this case, boronic pinacol esters had to be used instead of boronic acids owing to the inherent instability of pyrroles.^[24] Surprisingly, unlike the other heterocycles, pyrroles afforded considerably less reductive arylation product (Table 7). With the unprotected pyrrole **27a**, the reaction proceeded with poor



Scheme 3. Initial observations of deuterium incorporation.

Table 5. Chemodivergence between ring-opening and reductive arylation with variation in ligand.^[a]

Entry	Ligand	R	R'	Yield 2 [%] ^[b]	Yield 3 [%] ^[b]
1	J2	Cy	<i>t</i> -Bu	84	Trace
2	J1		<i>t</i> -Bu	70	20
3	J3		<i>t</i> -Bu	45	37
4	W3		Cy	40	35
5 ^[c]	J4			10	40
6 ^[d]	W1			11	26
7 ^[e]	W1			14	89
8	W2			trace	51

^[a] Reaction conditions as described in Table 1.

^[b] All yields are isolated.

^[c] 48% yield of **23**.

^[d] 61% yield of **23**.

^[e] PhB(OH)₂ and diazabicyclic **1d** were added *via* syringe pump over 1 h.

conversion, likely a result of incompatibility of the rhodium catalyst with most free amines (20% combined yield, entry 1).

The electron-withdrawing or -donating ability of the N-protecting group had a moderate influence on

chemoselectivity. Overall, rhodium migration was faster in pyrroles with electron-deficient protecting groups **27d** and **27e**. Thus more reductive arylation occurred with acylated and Boc-protected pyrrolyl-boronic esters, but in all cases the ring-opened product **28** was favoured with *t*-Bu-Josiphos (entries 4 and 5, Table 7).

A telling influence of steric effects was seen during deuterium quench experiments (Scheme 5). The reaction was performed in a THF/D₂O mixture, and deuterium incorporation was quantified in the reductive arylation products of the Boc-protected and the unprotected pyrroles (**29e** and **29a** respectively). In the case of the Boc group **29e**, deuteration occurred primarily at the β position, whereas in the unprotected example, **29a**, deuterium was observed only at the α position. This strong steric effect may explain the lack of reductive arylation in pyrroles in comparison to

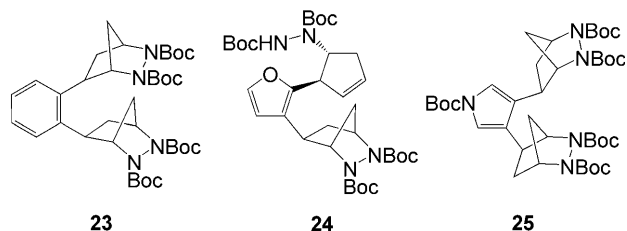
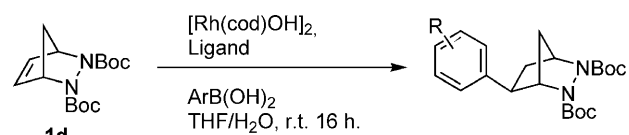
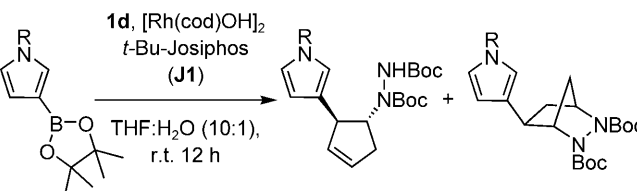
**Scheme 4.** Polysubstituted compounds derived from repeated carboration of substrate. **24** was formed using **J1** ligand. **23** and **25** were formed using **W1** ligand.

Table 6. Enantioselective reductive arylation of diazabicycle **1d** with substituted boronic acids.^[a]

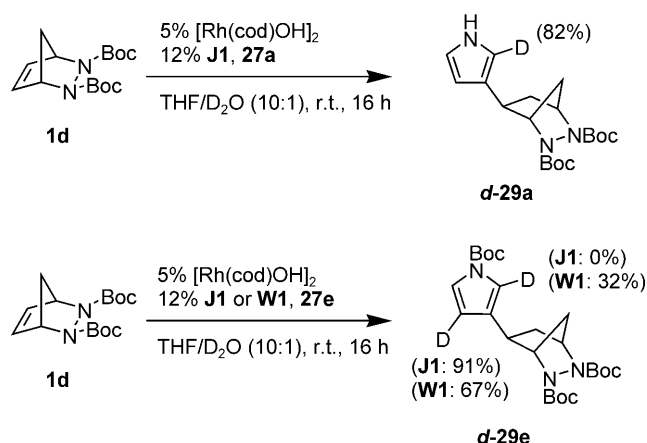
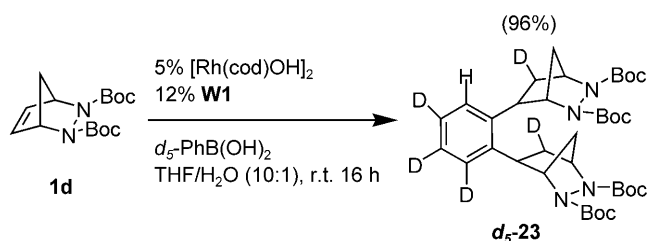
					
Entry	Ligand	Ar	Product	Yield [%] ^[b]	ee [%] ^[c]
1	W1		3d	69	62
2	J1		21	49	99
3	J1		26	47	99
4	W1		22	87	65
5	W2		29a	45	78
6	J1		20	66	99

^[a] Reaction conditions as in Table 1.^[b] All yields are isolated.^[c] Enantioselectivity was determined by chiral HPLC.**Table 7.** Selectivity between ring-opening and reductive arylation with pyrrolylboronic esters bearing different protecting groups.^[a]

					
Entry	R	28	ee [%] ^[c]	29	ee [%] ^[c]
		Yield [%] ^[b]		Yield [%] ^[b]	
1	H (a)	15	12	5	22
2	Me (b)	80	10	trace	–
3	TIPS (c)	87	30	11	75
4	Ac (d)	65	46	22	86
5	Boc (e)	68	37	30	80

^[a] Reaction conditions as in Table 1.^[b] All yields are isolated.^[c] Enantioselectivity was determined by chiral HPLC.

other heterocycles. Although the α position carries the more activated C–H bond, the rhodium insertion is hindered by steric bulk of the protecting group.

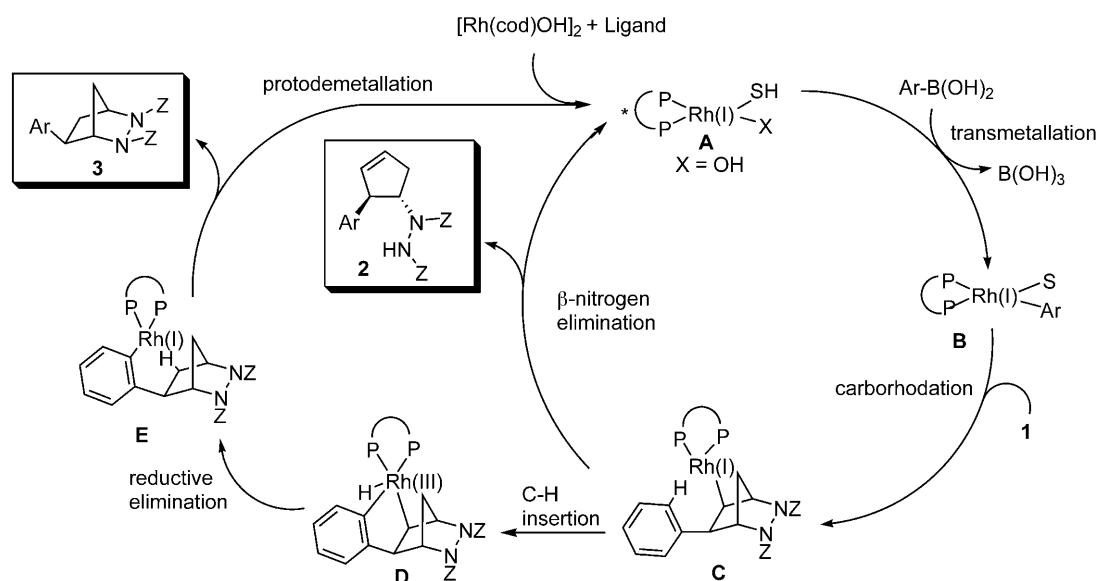
**Scheme 5.** Deuterium quench experiments conducted with pyrrolyl- and *N*-Boc-pyrrolylboronic esters.**Scheme 6.** Deuterium transfer experiment.

With Josiphos (**J1**) deuterium was incorporated exclusively at the β position, while with Walphos (**W1**) some deuteriation was seen at the α position as well. This discrepancy demonstrates that, although similar, the two ligands have distinctly different steric demands.

Mechanistic Aspects

The proposed mechanism for the reaction is illustrated in Scheme 7. Initially, transmetalation with the boronic acid gives rise to an arylrhodium species that coordinates to the alkene. The subsequent carboration is presumed to be irreversible and leads to formation of **C**. Complex **C** can ring-open to give product **2** or undergo a 1,4-rhodium migration to give arylrhodium intermediate **E**, which undergoes protodemetalation to yield **3**. Similar 1,4-migration of rhodium has been reported previously: notably in sequential C–H activation and quench observed by Miura,^[23] and in cyclization reactions reported by Hayashi,^[25] Iwasawa,^[26] and Murakami.^[27]

Chemodivergence arises following formation of complex **C**. Thereafter, the ring-opening and C–H activation are in competition. Our results show that outside of ligand control, rhodium 1,4-migration can be favoured by: 1) using electron-rich protecting groups



Scheme 7. Proposed mechanism accounting for the formation of **2** and **3**.

on hydrazine to reduce β -nitrogen elimination (Table 3); 2) using coordinating solvents to stabilize Rh(III) intermediate **D** (Table 2 and 4); and 3) utilizing σ -withdrawing groups on the aryl moiety to facilitate C–H insertion (Table 4).

The formation of hydridorhodium intermediate **D** is supported by the observation of deuterium transfer between the aromatic position and the diazabicyclo when deuterated phenylboronic acid was used (Scheme 6). Moreover deuterium experiments confirmed the validity of intermediate **E**. Conducting the reaction in a THF:D₂O mixture afforded *ortho*-deuterated product. These results were found to be consistent with several boronic acids (Scheme 3 and Scheme 5).

An intriguing observation was made in the pyrrolylboronic ester study. The enantioselectivities of **28** and **29** were not the same even though the two products likely form *via* a common intermediate **C** (Table 7). While initially perplexing, this observation shows that carborhodation precedes the rate-determining steps either for one or both of the products. The formation of complex **C** determines the relative stereochemistry of the molecule but not the enantiomeric excess of the products, which instead is a function of the rates of 1,4-migration and ring-opening.

Conclusions

In summary, a rhodium-catalyzed chemodivergent desymmetrization of diazabicycles with boronic acids was developed. Competing ring-opening and reductive arylation mechanisms gave access to substituted cyclopentenes **2** and diazabicycles **3**. Notably, ring-

opening proceeded in high yield and *ee* with electron-rich *ortho*-substituted boronic acids. Ligand control over the two competing reaction pathways was observed, where electron-deficient Walphos-type ligands favoured the 1,4-migration leading to **3**. Alternative factors in chemoselectivity were shown to be electronics and sterics of the boronic acids, the nature of the N-protecting group and the solvent used. Synthetic applications of the observed C–H insertion are currently under investigation.

Experimental Section

General Remarks

Unless otherwise specified, all experimental procedures were carried out under an argon atmosphere. ¹H and ¹³C NMR spectra were obtained at 400 MHz and 100 MHz, or 500 MHz and 125 MHz, respectively. Enantiomeric excess was determined by HPLC using Chiralcel AD or Chiralpak OD-H columns from Chiral Technologies. Boronic acids were purchased from commercial sources or synthesized from aryl halides according to a literature procedure.^[28] Bicyclic hydrazines were prepared by Diels–Alder cycloaddition.^[13] [Rh(cod)OH]₂ was conveniently prepared from [Rh(cod)Cl]₂ following a literature procedure.^[29] Compounds were purified with Silicycle™ Ultra-Pure 230–400 mesh silica gel.

Representative Procedure for the Rhodium-Catalyzed Desymmetrization of Diazabicyclo-[2.2.1]heptene (**1**)

To a 1-dram vial equipped with a septum and magnetic stir bar was added [Rh(cod)OH]₂ (3.1 mg, 0.0067 mmol) and *tert*-Bu-Josiphos (**J1**) (8.8 mg, 0.016 mmol). The vial was

purged with argon, then distilled THF (1.0 mL) and deionized water (0.2 mL) was added. The clear orange solution was stirred at room temperature for 15–20 min. Diazabicyclo[2.2.1]heptene dicarbamate **1d** (40 mg, 0.135 mmol) and phenylboronic acid (0.27 mmol, 2.0 equiv) were added together as a solution in distilled THF (1.0 mL), and the darkening reaction mixture was allowed to react at room temperature. After 16 h, TLC showed full consumption of **1d** (20% EtOAc/hexane; acidic vanillin stain). The reaction mixture was filtered on a short silica gel pad (~2 g), washing with four portions of ethyl acetate. The filtrate was concentrated under reduced pressure and purified by column chromatography (10% EtOAc/hexane). The ring-opened hydrazide 1,2-*trans*-*N,N'*-(2-phenylcyclopent-3-enyl)-di-(*tert*-butyl)hydrazine dicarboxylate (**2d**) was recovered as a colourless oil; yield: 43 mg (85%). The enantiomeric excess was determined on the purified product (*vide infra*). ¹H NMR (400 MHz, CDCl₃): δ = 7.31–7.17 (5H, m), 6.26 and 6.06 (1H, coalescing br.s), 5.89–5.83 (1H, m), 5.74–5.66 (1H, br.m), 4.70 (1H, br.s), 3.95 (br.s), 2.72–2.51 (2H, br.m), 1.55–1.05 (18H, br.m); ¹³C NMR (100 MHz, CDCl₃): δ = 156.0, 155.1, 143.9, 132.9 (2), 130.2 (2), 128.6, 127.8, 126.6, 81.4 (2), 68.2 and 66.4 (br, rotamers), 54.0, 35.4, 28.4 (3), 28.2 (3); IR (NaCl, neat film): ν = 3263 (br), 3056, 3025, 2977, 2930, 2865, 1744–1700 (br.), 1493, 1453, 1392, 1366, 1283, 1249, 1155, 1049, 952, 757, 699 cm⁻¹; MS (ESI): *m/z* (% rel. intensity) = 375.2 (MH⁺, 5), 319 (3), 297 (5), 263 (15), 219 (35), 143 (100), 128 (5); HR-MS (ESI): *m/z* = 375.2294, calcd. for C₂₁H₃₁N₂O₄ [MH⁺]: 375.2278; [α]_D^{24.3}: –75° (c 1.33, CHCl₃) for an 81.2:18.8 *er*, as determined by HPLC analysis: (Chiralcel AD, gradient 5–10% *i*-PrOH/hexane, 0.80 mL min⁻¹, 220 nm): t_R = 39.2 min [major], t_R = 42.7 min [minor]. The absolute configuration was assigned by derivatization to the unprotected amine and comparison to literature optical rotation values.^[3]

Supporting Information

Complete experimental procedures and spectroscopic data for compounds **7**, **8**, **17**, **18**, **23–26**, **27b**, **27d**, **28** and **29** are available as Supporting Information.

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